EXERCISE 4

Characteristic Polynomial • 15 IVPs Solved

# 15 Second-Order Homogeneous Linear ODEs with Initial Conditions

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PHASE 1: STATEMENT & SETUP

Find the general solution of the following differential equations, and then the unique solution satisfying the specified initial conditions:

(i) $\ddot{x} - 3\dot{x} + 2x = 0,\ x(0) = 2, \dot{x}(0) = 6$

(ii) $y'' - 4y' + 4y = 0,\ y(0) = 0, y'(0) = 3$

(iii) $z'' - 4z' + 13z = 0,\ z(0) = 7, z'(0) = 42$

(iv) $\ddot{y} + \dot{y} - 6y = 0,\ y(0) = -1, y'(0) = 8$

(v) $\ddot{y} - 4\dot{y} = 0,\ y(0) = 13, \dot{y}(0) = 0$

(vi) $\ddot{\theta} + 4\theta = 0,\ \theta(0) = 0, \dot{\theta}(0) = 10$

(vii) $\ddot{y} + 2\dot{y} + 10y = 0,\ y(0) = 3, \dot{y}(0) = 0$

(viii) $2\ddot{z} + 7\dot{z} - 4z = 0,\ z(0) = 0, \dot{z}(0) = 9$

(ix) $\ddot{y} + 2\dot{y} + y = 0,\ y(0) = 0, \dot{y}(0) = -1$

(x) $\ddot{x} + 6\dot{x} + 10x = 0,\ x(0) = 3, \dot{x}(0) = 1$

(xi) $4\ddot{x} - 20\dot{x} + 21x = 0,\ x(0) = -4, \dot{x}(0) = -12$

(xii) $\ddot{y} + \dot{y} - 2y = 0,\ y(0) = 4, \dot{y}(0) = -4$

(xiii) $\ddot{y} - 4y = 0,\ y(0) = 10, \dot{y}(0) = 0$

(xiv) $y'' + 4y' + 4y = 0,\ y(0) = 27, y'(0) = -54$

(xv) $\ddot{y} + \omega^2 y = 0,\ y(0) = 0, \dot{y}(0) = 1$

PHASE 2: FUNDAMENTAL FORMULATION

For $a\ddot{u} + b\dot{u} + cu = 0$, the exponential ansatz $u(t) = e^{rt}$ yields the auxiliary equation:

$$ar^2 + br + c = 0, \qquad \Delta = b^2 - 4ac$$

- $\Delta > 0$: $u(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ (Distinct Real).
- $\Delta = 0$: $u(t) = (c_1 + c_2 t)e^{rt}$ (Repeated Real).
- $\Delta < 0$: $u(t) = e^{\rho t}(c_1 \cos \omega t + c_2 \sin \omega t)$ (Complex Conjugate $r = \rho \pm i\omega$).

Constants $c_1, c_2$ are uniquely determined by Cauchy conditions $u(0) = u_0, \dot{u}(0) = v_0$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION (ALL 15 ITEMS)

**(i)** $\ddot{x} - 3\dot{x} + 2x = 0$ **with** $x(0) = 2,\ \dot{x}(0) = 6$

Auxiliary eq: $r^2 - 3r + 2 = 0 \implies (r-1)(r-2) = 0 \implies r_1 = 1, r_2 = 2.$

General solution: $x(t) = c_1 e^t + c_2 e^{2t} \implies \dot{x}(t) = c_1 e^t + 2c_2 e^{2t}.$

Initial conditions: $c_1 + c_2 = 2, c_1 + 2c_2 = 6 \implies c_2 = 4,\ c_1 = -2.$

**Solution:** $\mathbf{x(t) = -2e^t + 4e^{2t}}$

**(ii)** $y'' - 4y' + 4y = 0$ **with** $y(0) = 0,\ y'(0) = 3$

Auxiliary eq: $r^2 - 4r + 4 = 0 \implies (r-2)^2 = 0 \implies r = 2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{2t} \implies y'(t) = (2c_1 + c_2 + 2c_2 t)e^{2t}.$

Initial conditions: $y(0) = c_1 = 0; y'(0) = 2(0) + c_2 = 3 \implies c_2 = 3.$

**Solution:** $\mathbf{y(t) = 3te^{2t}}$

**(iii)** $z'' - 4z' + 13z = 0$ **with** $z(0) = 7,\ z'(0) = 42$

Auxiliary eq: $r^2 - 4r + 13 = 0 \implies (r-2)^2 + 9 = 0 \implies r = 2 \pm 3i$ ($\rho = 2, \omega = 3$).

General solution: $z(t) = e^{2t}(c_1 \cos 3t + c_2 \sin 3t).$

Derivative: $z'(t) = e^{2t}\left[(2c_1 + 3c_2)\cos 3t + (2c_2 - 3c_1)\sin 3t\right].$

Initial conditions: $z(0) = c_1 = 7; z'(0) = 2(7) + 3c_2 = 42 \implies 3c_2 = 28 \implies c_2 = \frac{28}{3}.$

**Solution:** $\mathbf{z(t) = e^{2t}\left(7\cos 3t + \frac{28}{3}\sin 3t\right)}$

**(iv)** $\ddot{y} + \dot{y} - 6y = 0$ **with** $y(0) = -1,\ y'(0) = 8$

Auxiliary eq: $r^2 + r - 6 = 0 \implies (r+3)(r-2) = 0 \implies r_1 = 2, r_2 = -3.$

General solution: $y(t) = c_1 e^{2t} + c_2 e^{-3t} \implies \dot{y}(t) = 2c_1 e^{2t} - 3c_2 e^{-3t}.$

Initial conditions: $c_1 + c_2 = -1, 2c_1 - 3c_2 = 8 \implies 2c_1 - 3(-1 - c_1) = 5c_1 + 3 = 8 \implies c_1 = 1,\ c_2 = -2.$

**Solution:** $\mathbf{y(t) = e^{2t} - 2e^{-3t}}$

**(v)** $\ddot{y} - 4\dot{y} = 0$ **with** $y(0) = 13,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 - 4r = 0 \implies r(r-4) = 0 \implies r_1 = 0, r_2 = 4.$

General solution: $y(t) = c_1 + c_2 e^{4t} \implies \dot{y}(t) = 4c_2 e^{4t}.$

Initial conditions: $\dot{y}(0) = 4c_2 = 0 \implies c_2 = 0; y(0) = c_1 = 13.$

**Solution:** $\mathbf{y(t) = 13}$ (Steady constant equilibrium)

**(vi)** $\ddot{\theta} + 4\theta = 0$ **with** $\theta(0) = 0,\ \dot{\theta}(0) = 10$

Auxiliary eq: $r^2 + 4 = 0 \implies r = \pm 2i$ ($\rho = 0, \omega = 2$).

General solution: $\theta(t) = c_1 \cos 2t + c_2 \sin 2t \implies \dot{\theta}(t) = -2c_1 \sin 2t + 2c_2 \cos 2t.$

Initial conditions: $\theta(0) = c_1 = 0; \dot{\theta}(0) = 2c_2 = 10 \implies c_2 = 5.$

**Solution:** $\theta(\mathbf{t}) = \mathbf{5}\sin\mathbf{2t}$

**(vii)** $\ddot{y} + 2\dot{y} + 10y = 0$ **with** $y(0) = 3,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 + 2r + 10 = 0 \implies (r+1)^2 + 9 = 0 \implies r = -1 \pm 3i\ (\rho = -1, \omega = 3).$

General solution: $y(t) = e^{-t}(c_1\cos 3t + c_2 \sin 3t).$

Derivative: $\dot{y}(t) = e^{-t}\left[(3c_2 - c_1)\cos 3t - (3c_1 + c_2)\sin 3t\right].$

Initial conditions: $y(0) = c_1 = 3; \dot{y}(0) = 3c_2 - 3 = 0 \implies c_2 = 1.$

**Solution:** $\mathbf{y(t) = e^{-t}\left(3\cos 3t + \sin 3t\right)}$

**(viii)** $2\ddot{z} + 7\dot{z} - 4z = 0$ **with** $z(0) = 0,\ \dot{z}(0) = 9$

Auxiliary eq: $2r^2 + 7r - 4 = 0 \implies (2r-1)(r+4) = 0 \implies r_1 = \frac{1}{2}, r_2 = -4.$

General solution: $z(t) = c_1 e^{t/2} + c_2 e^{-4t} \implies \dot{z}(t) = \frac{1}{2}c_1 e^{t/2} - 4c_2 e^{-4t}.$

Initial conditions: $z(0) = c_1 + c_2 = 0 \implies c_2 = -c_1; \dot{z}(0) = \frac{1}{2}c_1 + 4c_1 = \frac{9}{2}c_1 = 9 \implies c_1 = 2, c_2 = -2.$

**Solution:** $\mathbf{z(t) = 2e^{t/2} - 2e^{-4t}}$

**(ix)** $\ddot{y} + 2\dot{y} + y = 0$ **with** $y(0) = 0,\ \dot{y}(0) = -1$

Auxiliary eq: $r^2 + 2r + 1 = 0 \implies (r+1)^2 = 0 \implies r = -1$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{-t} \implies \dot{y}(t) = (c_2 - c_1 - c_2 t)e^{-t}.$

Initial conditions: $y(0) = c_1 = 0; \dot{y}(0) = c_2 - 0 = -1 \implies c_2 = -1.$

**Solution:** $\mathbf{y(t) = -te^{-t}}$

**(x)** $\ddot{x} + 6\dot{x} + 10x = 0$ **with** $x(0) = 3,\ \dot{x}(0) = 1$

Auxiliary eq: $r^2 + 6r + 10 = 0 \implies (r+3)^2 + 1 = 0 \implies r = -3 \pm i\ (\rho = -3, \omega = 1).$

General solution: $x(t) = e^{-3t}(c_1\cos t + c_2 \sin t).$

Derivative: $\dot{x}(t) = e^{-3t}\left[(c_2 - 3c_1)\cos t - (c_1 + 3c_2)\sin t\right].$

Initial conditions: $x(0) = c_1 = 3; \dot{x}(0) = c_2 - 3(3) = 1 \implies c_2 = 10.$

**Solution:** $\mathbf{x(t) = e^{-3t}\left(3\cos t + 10\sin t\right)}$

**(xi)** $4\ddot{x} - 20\dot{x} + 21x = 0$ **with** $x(0) = -4,\ \dot{x}(0) = -12$

Auxiliary eq: $4r^2 - 20r + 21 = 0. \Delta = 400 - 336 = 64 = 8^2 \implies r = \frac{20 \pm 8}{8} \implies r_1 = \frac{7}{2}, r_2 = \frac{3}{2}.$

General solution: $x(t) = c_1 e^{3t/2} + c_2 e^{7t/2} \implies \dot{x}(t) = \frac{3}{2}c_1 e^{3t/2} + \frac{7}{2}c_2 e^{7t/2}.$

Initial conditions: $c_1 + c_2 = -4, \frac{3}{2}c_1 + \frac{7}{2}(-4 - c_1) = -2c_1 - 14 = -12 \implies -2c_1 = 2 \implies c_1 = -1,\ c_2 = -3.$

**Solution:** $\mathbf{x(t) = -e^{3t/2} - 3e^{7t/2}}$

**(xii)** $\ddot{y} + \dot{y} - 2y = 0$ **with** $y(0) = 4,\ \dot{y}(0) = -4$

Auxiliary eq: $r^2 + r - 2 = 0 \implies (r+2)(r-1) = 0 \implies r_1 = 1, r_2 = -2.$

General solution: $y(t) = c_1 e^{t} + c_2 e^{-2t} \implies \dot{y}(t) = c_1 e^{t} - 2c_2 e^{-2t}.$

Initial conditions: $c_1 + c_2 = 4, c_1 - 2c_2 = -4 \implies 3c_2 = 8 \implies c_2 = \frac{8}{3},\ c_1 = \frac{4}{3}.$

**Solution:** $\mathbf{y(t) = \frac{4}{3}e^{t} + \frac{8}{3}e^{-2t}}$

**(xiii)** $\ddot{y} - 4y = 0$ **with** $y(0) = 10,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 - 4 = 0 \implies r = \pm 2.$

General solution: $y(t) = c_1 e^{2t} + c_2 e^{-2t} \implies \dot{y}(t) = 2c_1 e^{2t} - 2c_2 e^{-2t}.$

Initial conditions: $c_1 + c_2 = 10, 2(c_1 - c_2) = 0 \implies c_1 = c_2 = 5.$

**Solution:** $\mathbf{y(t) = 5e^{2t} + 5e^{-2t} = 10\cosh(2t)}$

**(xiv)** $y'' + 4y' + 4y = 0$ **with** $y(0) = 27,\ y'(0) = -54$

Auxiliary eq: $r^2 + 4r + 4 = 0 \implies (r+2)^2 = 0 \implies r = -2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{-2t} \implies y'(t) = (c_2 - 2c_1 - 2c_2 t)e^{-2t}.$

Initial conditions: $y(0) = c_1 = 27; y'(0) = c_2 - 2(27) = -54 \implies c_2 = 0.$

**Solution:** $\mathbf{y(t) = 27e^{-2t}}$

**(xv)** $\ddot{y} + \omega^2 y = 0$ **with** $y(0) = 0,\ \dot{y}(0) = 1$ **(** $\omega \neq 0$ **)**

Auxiliary eq: $r^2 + \omega^2 = 0 \implies r = \pm i\omega.$

General solution: $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t) \implies \dot{y}(t) = -c_1\omega\sin(\omega t) + c_2\omega\cos(\omega t).$

Initial conditions: $y(0) = c_1 = 0; \dot{y}(0) = c_2\omega = 1 \implies c_2 = \frac{1}{\omega}.$

**Solution:** $\mathbf{y(t) = \frac{1}{\omega}\sin(\omega t)}$

PHASE 4: MASTER SUMMARY TABLE

| Item | Differential Equation | Initial Conditions | Roots $r$ | Unique IVP Solution |
|---|---|---|---|---|
| (i) | $\ddot{x} - 3\dot{x} + 2x = 0$ | $x(0) = 2, \dot{x}(0) = 6$ | $1, 2$ | $\mathbf{-2e^{t} + 4e^{2t}}$ |
| (ii) | $y'' - 4y' + 4y = 0$ | $y(0) = 0, y'(0) = 3$ | $2$ (mult. 2) | $\mathbf{3te^{2t}}$ |
| (iii) | $z'' - 4z' + 13z = 0$ | $z(0) = 7, z'(0) = 42$ | $2 \pm 3i$ | $\mathbf{e^{2t}(7\cos 3t + \frac{28}{3}\sin 3t)}$ |
| (iv) | $\ddot{y} + \dot{y} - 6y = 0$ | $y(0) = -1, y'(0) = 8$ | $2, -3$ | $\mathbf{e^{2t} - 2e^{-3t}}$ |
| (v) | $\ddot{y} - 4\dot{y} = 0$ | $y(0) = 13, \dot{y}(0) = 0$ | $0, 4$ | $\mathbf{13}$ |
| (vi) | $\ddot{\theta} + 4\theta = 0$ | $\theta(0) = 0, \dot{\theta}(0) = 10$ | $\pm 2i$ | $\mathbf{5\sin 2t}$ |
| (vii) | $\ddot{y} + 2\dot{y} + 10y = 0$ | $y(0) = 3, \dot{y}(0) = 0$ | $-1 \pm 3i$ | $\mathbf{e^{-t}(3\cos 3t + \sin 3t)}$ |
| (viii) | $2\ddot{z} + 7\dot{z} - 4z = 0$ | $z(0) = 0, \dot{z}(0) = 9$ | $\frac{1}{2}, -4$ | $\mathbf{2e^{t/2} - 2e^{-4t}}$ |
| (ix) | $\ddot{y} + 2\dot{y} + y = 0$ | $y(0) = 0, \dot{y}(0) = -1$ | $-1$ (mult. 2) | $\mathbf{-te^{-t}}$ |
| (x) | $\ddot{x} + 6\dot{x} + 10x = 0$ | $x(0) = 3, \dot{x}(0) = 1$ | $-3 \pm i$ | $\mathbf{e^{-3t}(3\cos t + 10\sin t)}$ |
| (xi) | $4\ddot{x} - 20\dot{x} + 21x = 0$ | $x(0) = -4, \dot{x}(0) = -12$ | $\frac{3}{2}, \frac{7}{2}$ | $\mathbf{-e^{3t/2} - 3e^{7t/2}}$ |
| (xii) | $\ddot{y} + \dot{y} - 2y = 0$ | $y(0) = 4, \dot{y}(0) = -4$ | $1, -2$ | $\mathbf{\frac{4}{3}e^{t} + \frac{8}{3}e^{-2t}}$ |
| (xiii) | $\ddot{y} - 4y = 0$ | $y(0) = 10, \dot{y}(0) = 0$ | $\pm 2$ | $\mathbf{10\cosh(2t)}$ |
| (xiv) | $y'' + 4y' + 4y = 0$ | $y(0) = 27, y'(0) = -54$ | $-2$ (mult. 2) | $\mathbf{27e^{-2t}}$ |
| (xv) | $\ddot{y} + \omega^2 y = 0$ | $y(0) = 0, \dot{y}(0) = 1$ | $\pm i\omega$ | $\mathbf{\frac{1}{\omega}\sin(\omega t)}$ |

Study notes written from the official course material (ProblemsCh3.pdf; Robinson (2004).pdf). Online version: subjects/advanced-maths/problemas/topic-3-problems.html